

Renal depletion of *myo*-inositol is associated with its increased degradation in animal models of metabolic disease

H.-H. Chang, H.-N. Chao, C. S. Walker, S.-Y. Choong, A. Phillips, and K. M. Loomes

School of Biological Sciences and Maurice Wilkins Centre for Molecular Biodiscovery, University of Auckland, Auckland, New Zealand

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Chang HH, Chao HN, Walker CS, Choong SY, Phillips A, Loomes KM. Renal depletion of *myo*-inositol is associated with its increased degradation in animal models of metabolic disease. *Am J Physiol Renal Physiol* 309: F755–F763, 2015. First published August 26, 2015; doi:10.1152/ajprenal.00164.2015.—Renal depletion of *myo*-inositol (MI) is associated with the pathogenesis of diabetic nephropathy in animal models, but the underlying mechanisms involved are unclear. We hypothesized that MI depletion was due to changes in inositol metabolism and therefore examined the expression of genes regulating de novo biosynthesis, reabsorption, and catabolism of MI. We also extended the analyses from diabetes mellitus to animal models of dietary-induced obesity and hypertension. We found that renal MI depletion was pervasive across these three distinct disease states in the relative order: hypertension (−51%) > diabetes mellitus (−35%) > dietary-induced obesity (−19%). In 4-wk diabetic kidneys and in kidneys derived from insulin-resistant and hypertensive rats, MI depletion was correlated with activity of the MI-degrading enzyme *myo*-inositol oxygenase (MIOX). By contrast, there was decreased MIOX expression in 8-wk diabetic kidneys. Immunohistochemistry localized the MI-degrading pathway comprising MIOX and the glucuronate-xylulose (GX) pathway to the proximal tubules within the renal cortex. These findings indicate that MI depletion could reflect increased catabolism through MIOX and the GX pathway and implicate a common pathological mechanism contributing to renal oxidative stress in metabolic disease.

myo-inositol; MIOX; hypertension; diabetes; insulin resistance; kidney; obesity; inositol

CHRONIC KIDNEY DISEASE IS a serious disease that greatly impairs patients' health and life expectancy and affects 8–16% of population worldwide (16). It is characterized by a reduction in renal filtration and albuminuria and in advanced forms of the disease can lead to end-stage renal failure requiring kidney transplantation. Diabetic and hypertensive kidney diseases are the leading causes of end-stage kidney failure with a high mortality and morbidity accounting for >65% of total cases (9). The strikingly high prevalence of chronic kidney disease is projected to rise due to the growing epidemic of diabetes and hypertension.

Historically, the polycyclitol compound *myo*-inositol (MI) has attracted interest as its selective depletion in the renal cortex of the diabetic kidney in animal models has been implicated in the pathogenesis of diabetic complications (10, 12). MI functions as an organic osmolyte that enables renal cells to constantly adapt to hyperosmotic environments. MI and its subsequent phosphate derivatives are also involved in signaling pathways and membrane trafficking mechanisms

through phosphoinositide metabolism (10). In diabetes, MI depletion was proposed to arise as a consequence of increased activity of the polyol pathway whereby glucose is first converted to sorbitol by aldose reductase and then to fructose by sorbitol dehydrogenase. This “sorbitol-*myo*-inositol hypothesis” proposed that elevated intracellular sorbitol levels were ultimately responsible for MI depletion and impaired phosphoinositide metabolism (12).

Nevertheless, this mechanism is controversial as long-term supplementation with either an aldose reductase inhibitor to reduce sorbitol levels or replacement therapy with MI in the Cohen diabetic (type 2) rat had no measureable effects on the development of renal pathological changes (8). Similarly, renal depletion of MI persisted despite normalization of sorbitol levels in diabetic rats treated with an aldose reductase inhibitor, strongly suggesting that MI depletion occurs independently of the polyol pathway (29). The origin of inositol depletion in diabetes mellitus is therefore currently poorly understood as are the mechanisms linking it to the pathogenesis of renal complications.

We hypothesized that renal MI depletion in diabetes was due to changes in metabolic processes controlling inositol homeostasis. The kidney is a major site for de novo MI biosynthesis, reabsorption, and degradation (Fig. 1) (12). MI can be synthesized de novo from glucose-6-phosphate by MI-1-phosphate synthase (MIPS; EC 5.5.1.4) to produce MI-1-phosphate (26). The resulting MI-1-phosphate is subsequently dephosphorylated to produce free MI by the enzyme MI monophosphatase (IMPase; EC 3.1.3.25). MI is also selectively reabsorbed into the proximal tubules from the glomerular filtrate through inositol transporters, including sodium-dependent MI transporter subtypes 1 (SMIT1) and 2 (SMIT2) (2, 3, 20). Finally, MI can be degraded to D-glucuronate by the kidney-specific enzyme *myo*-inositol oxygenase (MIOX) and the glucuronate-xylulose (GX) pathway.

In principle, alterations in any of these metabolic components could affect intracellular MI levels. In the present study, we report on the expression of genes regulating MI de novo synthesis and its degradation in diabetes mellitus. Of particular interest was the enzyme MIOX, which metabolizes MI and is implicated in the pathogenesis of renal complications (11, 38, 40). We surmised that MI depletion might exist in other perturbed metabolic states other than diabetes, where impaired renal function also occurs. We therefore extended the experimental analyses to include animal models of insulin resistance and hypertension.

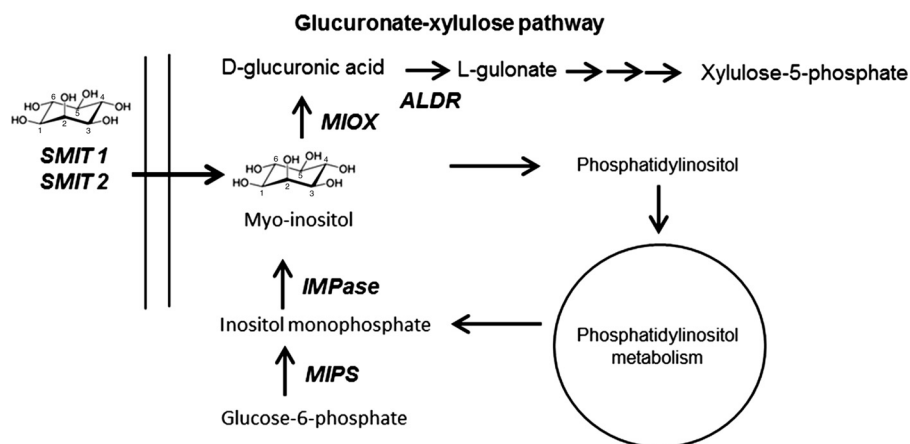
MATERIALS AND METHODS

Animal Models

Animal studies were approved by the University of Auckland Animal Ethics Committee. Animals were dual-housed under a con-

Address for reprint requests and other correspondence: K. Loomes, School of Biological Sciences and Maurice Wilkins Centre, Univ. of Auckland, Auckland 1010, New Zealand (e-mail: k.loomes@auckland.ac.nz).

Fig. 1. Renal *myo*-inositol (MI) metabolism. MI intracellular concentrations are determined by cellular uptake through inositol transporters, endogenous synthesis via glucose-6-phosphate, and degradation via the glucuronate pathway comprising the following steps: D-glucuronate → L-gulonate → 3-keto-gulonate → L-xylulose → xylitol → D-xylulose → xylulose-5-phosphate (34). *Myo*-inositol phosphate synthase (MIPS), inositol monophosphatase (IMPase), sodium *myo*-inositol transporter 1 (SMIT1); sodium *myo*-inositol transporter 2 (SMIT2); *myo*-inositol oxygenase (MIOX); aldehyde reductase (ALDR). Adapted from Ref. 12 with permission.



trolled environment (12:12-h light-dark cycle, 50–70% humidity, room temperature $22 \pm 2^\circ\text{C}$) with ad libitum access to food (standard rodent chow, Teklad, IN, unless otherwise specified) and tap water.

Streptozotocin-induced diabetes. Seven- to eight-week-old male Wistar rats, weighing between 250 and 300 g, were injected intravenously with 50 mg/kg body wt streptozotocin (STZ; Sigma, St Louis, MO). Blood glucose levels were monitored following days 1 and 3 postinjection (Accu-Chek). Animals were kept for 4 wk postinjection before harvesting of kidneys.

Dietary-induced obesity. Six-week-old C57BL/6 male mice were fed a control diet (fat content as 10% of total energy, 12450B, Research Diets, New Brunswick, NJ) or a high-fat diet (fat content as 60% of total energy, 12492, Research Diets) for 6 mo as a model for insulin resistance and obesity as reported by us previously (36).

Hypertension. The blood pressure of 12-mo-old male spontaneously hypertensive rats (SHR) and Wistar-Kyoto (WKY) rats were monitored indirectly by tail cuff (MC4000 Blood Pressure Analysis System, Hatteras Instruments).

Immunohistochemistry

Whole rat kidneys were sliced in half longitudinally after excision and fixed in 4% phosphate-buffered paraformaldehyde fixative overnight (12–18 h). Kidney pieces were washed three to four times in 50% ethanol and stored in cassettes in 70% ethanol. Samples were embedded in paraffin, and sections of 5- μm thickness were sliced and mounted onto slides. Slides were heated at 60°C in a tissue drying oven (TDO 50) for 3 min to dissolve the paraffin wax. They were then dewaxed in xylol for 4 min, rinsed in 100% ethanol, followed by 95% ethanol. Slides were washed in $1\times$ PBS for 15 min on a shaking platform before blocking in 5% donkey serum for 1 h at 37°C . They were then washed again in $1\times$ PBS for 15 min and incubated with rabbit polyclonal mouse anti-MIOX antibody (generated in-house) and goat polyclonal anti-AKR1A1 (aldehyde reductase) primary antibodies (Abcam) for 2 h at room temperature. Slides were then washed in $1\times$ PBS for 15 min before incubation with secondary antibody solution containing a mixture of fluorescein (FITC)-conjugated donkey anti-rabbit IgG and rhodamine red-X-conjugated donkey anti-goat IgG (Jackson ImmunoResearch Laboratories) secondary antibodies for 2 h at room temperature in the dark. Fluorescent images were viewed using the (Axioskop plus, Zeiss) fluorescence microscope, using different filters (495-nm excitation, blue light for FITC and 560-nm excitation, green light for rhodamine red). Images were taken at $\times 4$ magnification with an AxioCamHR camera (Zeiss).

MI Quantitation in the Kidney

Whole kidneys were homogenized in water in the presence of 10 nmol of deuterated MI as an internal standard. Samples were deproteinized by adding an equal volume of ethanol and then freeze-dried

overnight (SP Industries). Inositols present within the homogenates were converted to trimethylsilyl (TMS) derivatives and analyzed by gas chromatography/mass spectrometry (GC/MS). The peak intensity of MI was normalized against the peak of deuterated MI.

Real-Time PCR

Kidney total RNA was extracted with TriPure Reagent (Roche). Total RNA was DNase-digested with DNA-free (Ambion) before cDNA construction with a Transcriptor First Strand cDNA synthesis kit (Roche) using random hexamer primers. cDNA samples were diluted fivefold, and 1 μl of cDNA was mixed with 1 μl of both forward and reverse primers (1.25 μM), 5 μl of SYBR green master mix (Roche), and 2 μl of PCR-grade water in a LightCycler 480 Multiwell Plate 384 (Roche). Primer amplification efficiency was determined from the standard curve from three independent quantitative (q) PCR runs. Primer pair amplification efficiencies <1.8 were discarded as indicative of low signal-to-noise ratio (17).

Gene expression was normalized to the geometric mean of a combination of reference genes selected from seven common housekeeping genes using GeNorm software (Primer Design). The candidate housekeeping genes included β -actin (ACTB), ubiquitin C (UBC), tyrosine 3-monooxygenase/tryptophan 5-monooxygenase activation protein, zeta polypeptide (YWHAZ), 18S rRNA (18S), hypoxanthine-guanine phosphoribosyltransferase (HPRT), and hydroxymethylbilane synthase (HMBS). Primers are listed in Table 1, with the specificity of all primer pairs validated using two independent approaches. First, cDNA derived from RNA isolated from Wistar rat kidneys was used as a template for conventional PCR. PCR products were separated on a 2% agarose gel and stained with SYBR Safe (Invitrogen). All primer pairs gave a single band of the expected size. The identities of the PCR products were confirmed by the sequencing reactions. Second, the specificity of each primer pair was validated by performing a dissociation curve using the LightCycler 480 (Roche).

Table 1. Primers for target and reference genes

Gene	Forward Primer	Antisense Primer
Target genes		
MIOX	TCCGAACTACACTTCAGGC	CAGATGGAACCACTCCTTGT
SMIT1	ATGTGCTGTTCATCCTGCTC	TTCTCTCAAACCTCCAACCT
SMIT2	GCTGCCTATGAGCTTAATGGCTTG	CCAGACCTCCTGCAACAGTG
IMP	GTGACCGAGTTGGGCTCTTCC	CTCCCGTTGCCACAAGGC
MIPS	GGSCCTAAGACCATGTCCATCG	GATSACCAACGAGTGTCWGG
Reference genes		
ACTB	GGGAAATCGTGCGTGACATT	GCGGCAGTGCGCATCTC
HPRT	GCGAAAGTGGAAAAGCCAAGT	GCCACATCAACAGGACTCTTGTAG
UBC	GGCAAGATCCAGGACAAGGAG	GCCTGCAAGATCAGCCTCT
YWHAZ	GGACATCGGATACCAAGGA	GGGACAGCATGGATGACAAA

See the text for definitions.

All primer pairs yielded a single peak without the presence of any secondary peaks.

The C_p value of each reference gene was converted into a relative ratio before the analysis. For each cDNA sample, a corresponding no-RT negative control was included for the determination of genomic DNA contamination. Samples were discarded if the difference between the C_p values of the sample and the no-RT control was less than eight as indicative of significant DNA contamination. Melting curves were generated for each primer pair to confirm the purity of the amplified products. Primer pairs with two peaks yielded from the melting curve were discarded. The experimental amplification efficiency of each primer pair was determined by generating a standard curve of C_p number against relative cDNA abundance. For each primer pair, three independent qPCR runs were performed and the average of the amplification efficiencies from all three runs was calculated.

Western Blotting

Whole kidney samples were homogenized in RIPA buffer [50 mM Tris-HCl, 150 mM NaCl, 1 mM EDTA, 0.1% (wt/vol) SDS, 1% (vol/vol) Triton X-100, 0.25% (wt/vol) deoxycholate] at 20 mg wet wt/ml by TissueLyzer II (Qiagen). Protein concentration was determined using a BCA assay (Sigma). Kidney lysates (20 μ g) were resolved on NuPage 4–12% gradient gels (Invitrogen), and separated proteins were transferred onto a polyvinylidene difluoride (PVDF) membrane. Membranes were blocked overnight with 5% nonfat milk in PBST [136.9 mM NaCl, 2.68 mM KCl, 12 mM Na_2HPO_4 , 1.38 mM KH_2PO_4 , 0.5% (vol/vol) Tween 20, pH 7.4]. Membranes were then incubated with rabbit anti-MIOX IgG (generated in-house) or mouse anti- β -actin IgG (Santa Cruz Biotechnology) followed by incubation with 655 QDot goat anti-rabbit IgG (Invitrogen) or 655 QDot goat anti-mouse IgG (Invitrogen), respectively. Band intensities were visualized using a Fujifilm detection system.

MIOX Activity

MIOX activity was determined using the modified orcinol assay (24). Kidney samples were homogenized in lysis buffer (20 mM sodium acetate, 1 mM Fe^{2+} , 2 mM cysteine, and 1 mM glutathione) and centrifuged at 10,000 g for 30 min. at 4°C. Sample lysates were incubated with assay buffer (50 mM sodium acetate, 1 mM Fe^{2+} , 2 mM cysteine, 60 mM MI) for 30 min. at 37°C. Reactions were terminated by heating the samples to 100°C for 5 min followed by centrifugation. Supernatants were mixed with orcinol reagent (28 mM orcinol, 3 mM Fe^{3+} in concentrated HCl) and incubated for 30 min. at 57°C. A blank for each sample (lysate incubated with assay buffer without MI) was included and incubated in parallel. Reactions were terminated by placing tubes in ice water for 5 min. and absorbance measured at 620 nm.

Statistical Analyses

All data are expressed as means $\pm$ SE. Statistical analyses were performed using GraphPad Prism version 5.02 (GraphPad Software, San Diego, CA) for an unpaired two-tailed Student's t -test and JMP version 8 (SAS Institute, Cary, NC) for two-way ANOVA. $P < 0.05$ was considered statistically significant.

RESULTS

MI Catabolism Occurs in the Proximal Tubules Through MIOX and the GX Pathway

In mammals, MIOX and the GX pathway are the only known route by which MI is catabolized (13). The GX pathway is also constitutively active in humans as demonstrated by the benign human genetic condition of essential pentosuria (13). The product of the MIOX reaction, D-glucuronate, is metabo-

lized by the GX pathway, which is active primarily in the liver and kidney (Fig. 1). Aldehyde reductase is the first enzyme of the GX pathway and reduces the product of the MIOX reaction, D-glucuronate, to L-gulonate. Its role in glucuronate metabolism was confirmed by its inhibition through in vivo administration of the inhibitor AL-1576 to rats, which increased urinary excretion of glucuronate sixfold (1). Its activity in the kidney is ~ 10 -fold higher than in the liver and intestine, representing $\sim 1\%$ of soluble protein in the kidney (1) consistent with a major metabolic function in this organ.

To localize the site for MI degradation in the kidney, we investigated the localization of MIOX and the GX pathway in non-diabetic rat kidney tissue using immunofluorescence microscopy (Fig. 2). Immunoreactivities for both MIOX (Fig. 2A) and the first enzyme in the GX pathway, aldehyde reductase, were observed in the renal cortex but not in the medulla or glomeruli (Fig. 2B, arrows). Immunofluorescence was not observed for MIOX (Fig. 2C) or aldehyde reductase (Fig. 2D) in control sections incubated with secondary antibodies (no primary antibody controls). The specificity of the immunofluorescence signal for MIOX was further confirmed by preabsorbance of anti-MIOX polyclonal antibodies with rat recombinant MIOX (5 μ g/ml), which resulted in attenuation of the fluorescence (Fig. 2E). These findings show that MIOX and aldehyde reductase and therefore MI degradation is localized to the proximal tubules.

It has been reported that MIOX and aldehyde reductase exist as part of a multiprotein enzyme complex (30). We therefore investigated whether stable protein interactions existed between these two enzymes. Batchwise fractionation of rat kidney lysates by anion exchange chromatography showed that MIOX eluted with 150 mM KCl as detected by Western blot analysis (Fig. 3). By comparison, detection of eluted fractions with anti-AKR1A1 antibody identified aldehyde reductase in the earlier wash fraction (Fig. 3). Further purification of the ion exchange fraction containing MIOX by gel filtration chromatography revealed an elution position consistent with a monomeric molecular mass (35 kDa) (data not shown). These experiments provided no evidence for coeluting species. We also performed various pulldown assays of crude rat kidney lysates utilizing different buffer conditions with either histidine-tagged recombinant rat MIOX, or by immunoprecipitation of endogenous proteins with MIOX- and aldehyde reductase-specific (anti-AKR1A1) antibodies. All these studies together with other crystallization studies with recombinant MIOX and aldehyde reductase provided no evidence that they form stable complexes (data not shown).

Renal MI Depletion in 4-Wk Diabetic Kidneys Is Associated with Increased MIOX Expression

Rodent models of diabetes mellitus have proven to be essential experimental tools to elucidate underlying pathological mechanisms (7). We utilized a STZ-induced rat model that produces a type 1 diabetes phenotype through selective pancreatic islet β -cell toxicity and insulin deficiency (33). We hypothesized that this particular model was suitable to investigate the effects of prolonged hyperglycemia on inositol metabolism in the absence of confounding metabolic effects due to systemic insulin resistance.

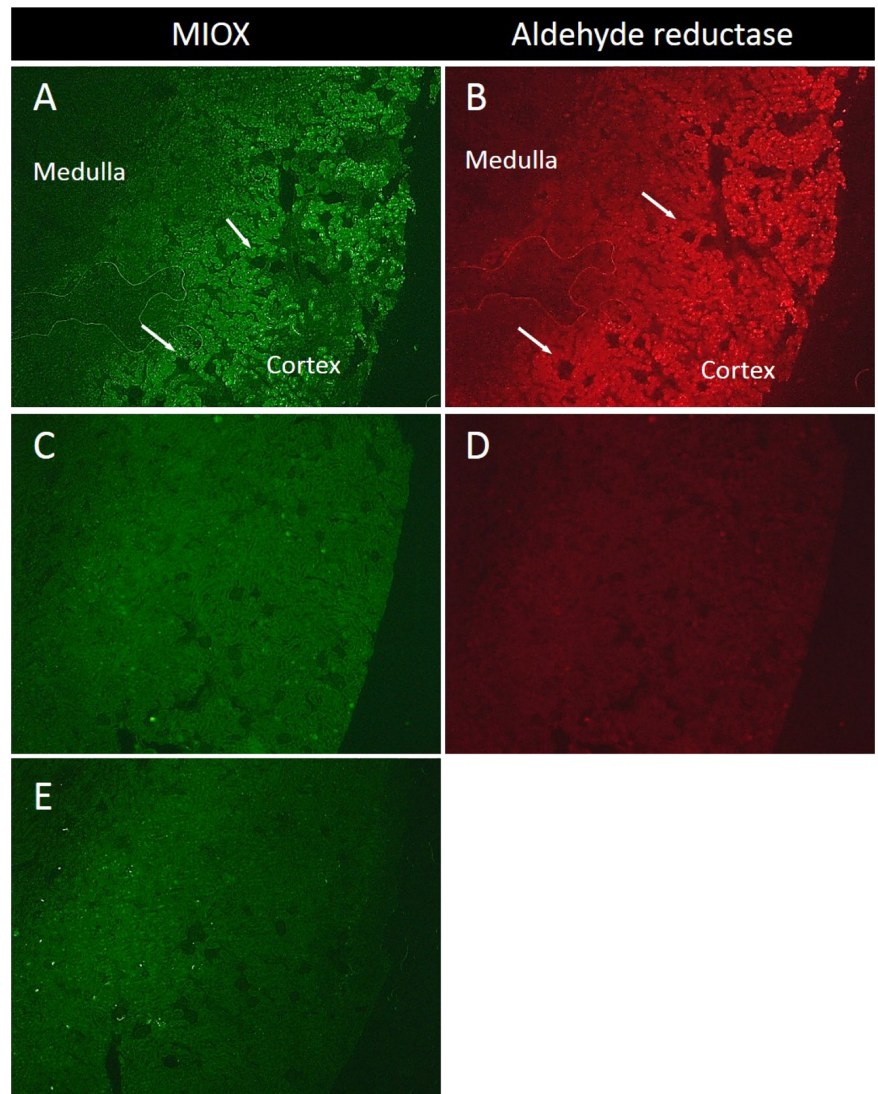


Fig. 2. Localization of MIOX and aldehyde reductase in longitudinal rat kidney sections. Double immunofluorescence staining of kidney sections for MIOX (A) and aldehyde reductase (B) show fluorescence localized to the renal cortex. Panels C and D show no primary antibody controls, respectively. E: preabsorption of MIOX polyclonal antibody with 5 µg/ml recombinant MIOX. All images were taken at $\times 4$ magnification.

For STZ-induced diabetes, male Wistar rats received a single injected dose of STZ, which resulted in a dramatic increase in blood glucose (>15 mM) on the first day postinjection. These glucose levels were maintained throughout (data not shown). At 4 wk, plasma glucose levels were elevated by about fivefold while body weight was reduced by 19% compared with age-matched nondiabetic controls ($n = 13$, $P < 0.001$) (Table 2).

Total RNAs from whole kidneys of 4- and 8-wk STZ-induced diabetic rats and sham controls were isolated and reverse transcribed into cDNA. For 4-wk diabetic kidneys, qPCR revealed increased MIOX gene expression compared with the corresponding nondiabetic controls by 34% ($n = 6$, $P < 0.05$) with no changes in expression of IMPase and MIPS or aldose reductase (Fig. 4A). We have shown previously that SMIT1 and SMIT2 were also significantly elevated in diabetic 4-wk diabetic kidneys compared with the corresponding nondiabetic controls by 34 and 48%, respectively (6). Increased MIOX expression in 4-wk diabetic kidneys was confirmed by a 25% increase in both protein expression ($n = 7$, $P < 0.01$) (Fig. 4B) and activity ($n = 6$, $P < 0.01$) (Fig. 4C). Corresponding analyses of renal MI content in whole kidney lysates

by GC/MS revealed a 35% decline ($n = 7$, $P < 0.01$) in 4-wk diabetic vs. nondiabetic kidneys (Fig. 4D).

By comparison, expression of MIOX and SMIT1 was significantly downregulated in 8-wk diabetic kidneys compared with the nondiabetic controls by 37 ($n = 9$, $P < 0.05$) and 43% ($n = 9$, $P < 0.05$), respectively (Fig. 4E). Otherwise, there were no changes in the gene expression of SMIT 2, IMPase, MIPS, or aldose reductase (Fig. 4E). Interestingly, MI depletion persisted in 8-wk diabetic kidneys compared with corresponding controls by 27% ($n = 7$, $P < 0.01$) (data not shown).

Renal MI Depletion and Increased MIOX Expression Occur in Animal Models of Dietary-Induced Obesity and Hypertension

We investigated two distinct disease phenotypes. The first was a diet-induced obesity animal model where C57BL/6 mice were fed a 60% fat diet for a period of 6 mo. We showed previously that these mice developed insulin resistance and displayed impaired metabolic health compared with C57BL/6 mice cohorts fed a control 10% fat diet (36). We kept kidney tissues banked from this study, which we were able to use here.

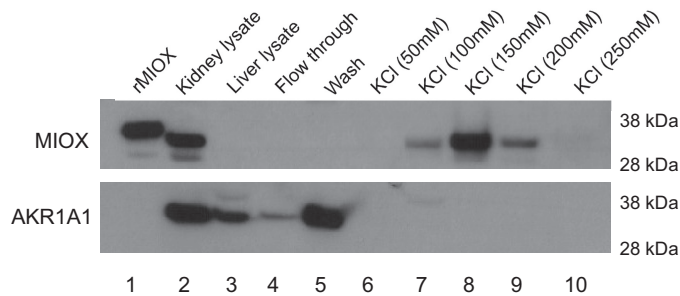


Fig. 3. Purification of MIOX and aldehyde reductase (AKR1A1) by anion exchange chromatography. Crude kidney lysate prepared in 25 mM Tris-HCl, pH 8.0, 50 mM KCl was purified on a HiTrap Q FF IEX column. Proteins were eluted with a stepwise KCl gradient. Fractions were analyzed for the presence of MIOX and aldehyde reductase by Western blot analysis. Samples were analyzed as follows: recombinant MIOX (rMIOX; 12 ng, lane 1; kidney lysate, 20 mg, lane 2); liver lysate (20 mg, lane 3); flow-through fraction (lane 4); wash fraction (lane 5); fractions eluted with 50 mM (lane 6), 100 mM (lane 7), 150 mM (lane 8), 200 mM (lane 9), and 250 mM (lane 10) KCl. The polyvinylidene difluoride membrane was probed with anti-MIOX antibody (top) then stripped and reprobed with anti-AKR1A1 antibody (bottom). Molecular weight markers are shown to the right. The higher molecular weight of rMIOX is consistent with the presence of the N-terminal 6-amino acid histidine tag (5).

The organs were harvested at the 6 month time point and immediately frozen at -80°C . Table 2 summarizes the phenotypes for the specific subset of these animals we used in this current analysis. Mice fed the high-fat diet displayed a 28% increase in body weight ($n = 12$, $P < 0.001$) compared with the control diet group. At the 6-mo time point, there was no difference in blood glucose levels between the two groups. High fat-fed mice displayed hyperinsulinemia with a 2.4-fold higher circulating serum insulin ($n = 10$, $P < 0.01$) together with impaired insulin sensitivity and glucose clearance as measured by intraperitoneal glucose tolerance tests and insulin tolerance tests. In our earlier study, we also showed that mice fed the 60% fat diet displayed fatty liver, altered adipokine profiles, and accumulation of triglyceride in skeletal muscle (36).

Analysis of the kidneys by qPCR revealed a 14% increase in SMIT2 gene expression in mice fed the high-fat diet compared with mice fed the 10% control diet (Fig. 5A). However, we failed to detect a corresponding increase in protein expression for SMIT2 by Western blotting (data not shown). MIOX gene expression was significantly elevated by 34% in the kidneys isolated from high fat-fed mice ($n = 7$, $P < 0.001$) with no changes in gene expression of IMPase and MIPS (Fig. 5A). MIOX protein expression was also increased by 20% ($n = 7$, $P < 0.01$) in kidneys isolated from the high-fat mice cohort (Fig. 5B) with a corresponding 14% increase ($n = 7$, $P < 0.05$) in MIOX enzymatic activity (Fig. 5C). The increased MIOX expression was associated with a 19% ($n = 6$, $P < 0.05$) reduction in MI content in whole kidney lysate in kidneys isolated from the high fat-fed group (Fig. 5D).

The second disease phenotype investigated was the hypertensive state. SHR are a strain of WKY rats where kidney impairment occurs (15). Normotensive WKY rats are commonly employed as the corresponding control. In this animal model, development of hypertension begins at 5–6 wk of age and reaches systolic pressures between 180 and 200 mmHg in adulthood. In the present study, 1-yr-old male SHR exhibited a

21% increase ($n = 10$, $P < 0.001$) in body weight compared with age-matched male WKY rats (Table 2). Both diastolic and systolic blood pressures were also significantly higher in the SHR rats compared with WKY rats by 21%, respectively ($n = 10$, $P < 0.001$).

qPCR revealed no changes in either the inositol transporters SMIT1 or SMIT2 in kidneys isolated from either WKY rats or SHR (Fig. 6A). By comparison, MIOX gene expression was significantly elevated by 140% in kidneys isolated from SHR ($n = 6$, $P < 0.01$) but again with no changes in gene expression of either IMPase or MIPS (Fig. 6A). Increased RNA expression in hypertensive kidneys was matched by a 45% increase in MIOX protein expression ($n = 7$, $P < 0.001$) (Fig. 6B) and a 2.5-fold increase in MIOX enzymatic activity ($n = 7$, $P < 0.05$) (Fig. 6C). Increased MIOX expression was associated with a 51% reduction in MI content ($n = 6$, $P < 0.01$) in hypertensive kidneys (Fig. 6D).

Overall, these findings showed that MI depletion was associated with increased MIOX expression in animal models of insulin resistance, hypertension, and diabetes mellitus at the 4-wk time point. The extent of MI depletion was correlated with MIOX activity that was highest for hypertension > 4-wk diabetes mellitus > insulin-resistant state (Fig. 6E).

DISCUSSION

Renal MI depletion has been reported previously in animal models of diabetes mellitus, although its significance in the pathogenesis of renal complications is unclear. Its origin has been proposed to occur as a consequence of hyperglycemia that increases sorbitol production through the polyol pathway (12). Our findings now show MI depletion is a persistent feature of hypertensive and insulin-resistant states and therefore can oc-

Table 2. Animal phenotypes for diabetes, dietary-induced obesity, and hypertension

STZ-Induced Diabetic Rats	Control	STZ
Body weight, g	435 $\pm$ 7	353 $\pm$ 11***
Blood glucose, mM	5.21 $\pm$ 0.11	27.84 $\pm$ 0.90***
Fat-fed C57BL mice	10% fat	60% fat diet
Body weight, g	42.7 $\pm$ 0.9	54.6 $\pm$ 0.9***
Blood glucose, mM	8.62 $\pm$ 0.28	8.67 $\pm$ 0.67
Serum insulin, ng/l	1,197 $\pm$ 158	2,840 $\pm$ 531***
ITT area under curve	8,676 $\pm$ 534	11,900 $\pm$ 556***
GTT area under curve	17,180 $\pm$ 465	21,300 $\pm$ 992***
Spontaneously hypertensive rats	WKY	SHR
Body weight, g	388 $\pm$ 9	473 $\pm$ 6***
Blood glucose, mM	6.73 $\pm$ 0.53	6.81 $\pm$ 0.71
Systolic blood pressure, mmHg	155 $\pm$ 3	187 $\pm$ 3***
Diastolic blood pressure, mmHg	140 $\pm$ 4	169 $\pm$ 6***

Values are means $\pm$ SE. For streptozotocin (STZ)-induced diabetes, animal body weights and blood glucose concentrations at 4 wk after intravenous injection of STZ or saline ($n = 13$ animals/group) are shown. For dietary-induced obesity, measurements were taken in C57BL/6 mice following 6 mo on control (10% fat) and high-fat (60%) diets ($n = 10$ –12 animals/group). Blood glucose levels were measured the day before euthanasia. The data shown are summarized from an animal subset of previously reported experimental data (36). For spontaneously hypertensive rats, measurements were taken at 1 yr ($n = 10$ animals/group). Statistical analysis were performed using Student's unpaired *t*-test. *** $P < 0.001$ vs. the respective control.

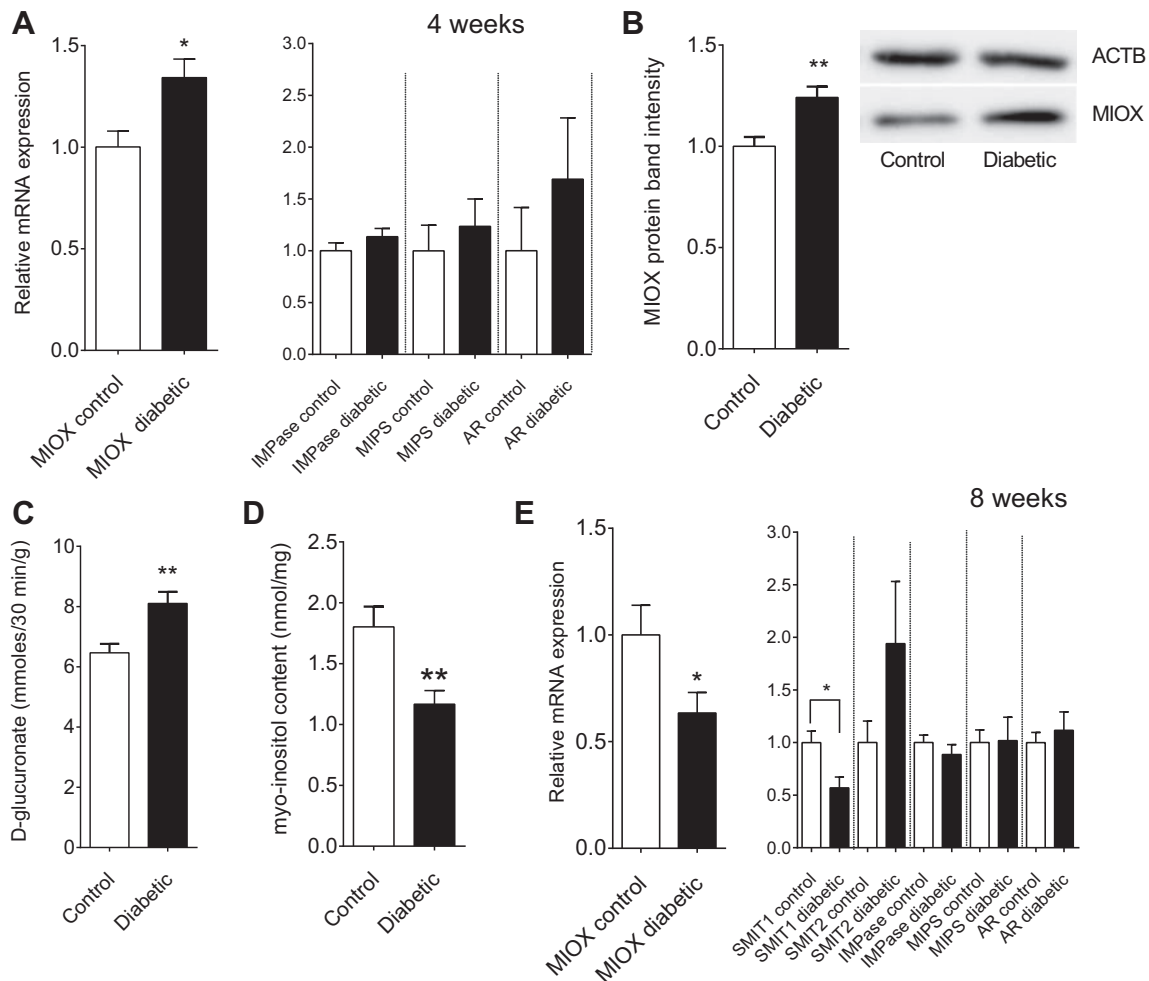


Fig. 4. MI depletion and gene expression profiles in streptozotocin (STZ)-diabetic rat kidneys. *A*: mRNA gene expression profiles in 4-wk postinjection rat kidneys ($n = 5-6$ kidneys/group). *B*: Western blot analysis of MIOX protein expression in 4-wk postinjection rat kidneys. Protein band intensities were normalized to β -actin (ACTB; $n = 7$ kidneys/group). *C*: MIOX activity in 4-wk postinjection rat kidneys measured as mmoles produced over 30 min/g kidney wet wt ($n = 8$ kidneys/group). *D*: intracellular MI concentrations in 4-wk postinjection rat kidneys normalized to kidney wet wt ($n = 7$ animals/group). *E*: mRNA gene expression profiles in 8-wk postinjection rat kidneys ($n = 5-6$ kidneys/group). Values are means $\pm$ SE. Statistical analysis was performed using Student's unpaired *t*-test. For *A* and *E*, mRNA expression levels were normalized to the geometric mean of the 3 reference genes, ACTB, HPRT, and UBC, and expressed as ratios relative to control kidneys. AR, aldose reductase. ** $P < 0.01$ vs. control. * $P < 0.05$ vs. control.

cur independently of hyperglycemia. In fact, MI depletion was most pronounced for the hypertensive phenotype. By comparison, we found no changes in gene expression of MIPS and IMPase, suggesting that inositol biosynthetic capacity was unaltered. The striking association with increased MIOX expression in kidneys from hypertensive, insulin resistant, and 4-wk diabetic animals indicates that MI degradation is likely a significant regulator of intracellular MI levels within the kidney.

The increased MIOX expression in 4-wk diabetic kidneys is consistent with high circulating glucose levels and the presence of carbohydrate-responsive elements in the 5' upstream gene sequence (27). Previous studies have also shown increased MIOX expression in animal models of diabetes mellitus (24, 39). MIOX expression is also sensitive to the GX pathway intermediate xylose and the final product, xylulose-5-phosphate, which activates carbohydrate-responsive element binding protein (ChREBP) (27). Increased metabolic activity through the GX pathway could therefore conceivably exert a positive feedback loop that sustains MIOX expression.

The mechanisms underlying increased MIOX expression in hypertensive and insulin-resistant states are less clear. However, the MIOX gene does also contain both osmotic- and oxidant-responsive elements (24). One possibility is that both hypertensive and insulin-resistant phenotypes promote perturbed metabolic states that drive MIOX expression through these mechanisms.

These findings suggest that increased MI degradation through the GX pathway occurs across all these distinct disease phenotypes at some phase of their respective time course. Of special interest is the implication for the GX pathway as a metabolic source of oxidative stress, which in turn contributes to fibrosis and progressive decline in renal function (35). The GX pathway contains enzymatic steps that increase cellular redox potential and also consume NADPH, thereby facilitating reactive oxygen species production and a weakening of cellular antioxidant defenses (13). Both hypertensive and diabetic nephropathies are characterized by oxidative stress components (4, 22, 31). Kidney impairment is also associated with metabolic syndrome, in which oxidative stress is a contributory

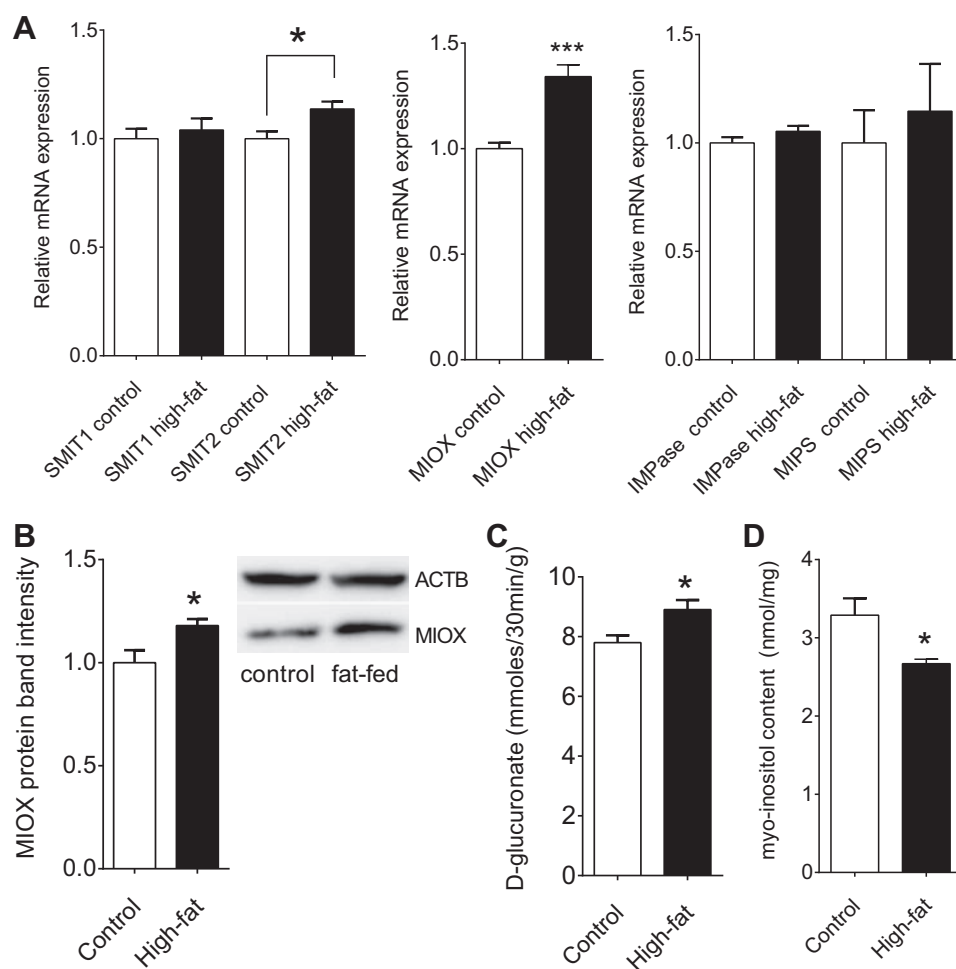


Fig. 5. MI depletion and gene expression profiles in kidneys from insulin-resistant C57/BL mice. **A:** mRNA gene expression profiles ($n = 7$ kidneys/group) normalized to the geometric mean of the reference genes, ACTB, UBC, YWHAZ and expressed as ratios relative to control kidneys. **B:** Western blot analysis of MIOX expression. Protein band intensities were normalized to ACTB ($n = 7$ kidneys/group). **C:** MIOX activity in kidney lysates measured as mmoles produced over 30 min/g kidney wet wt ($n = 6-7$ kidneys/group). **D:** intracellular MI concentration normalized to kidney wet wt ($n = 6$ animals/group). Values are means $\pm$ SE. Statistical analyses were performed using Student's unpaired *t*-test. *** $P < 0.001$ vs. control. * $P < 0.05$ vs. control.

component (28). An intriguing possibility is that increased MI catabolism through the GX pathway provides a common pathological mechanism contributing to oxidative stress in metabolic disease.

The involvement of MIOX and the GX pathway in oxidative stress is directly supported by studies in cultured cells showing that selective knockdown of MIOX expression attenuates hyperglycemia-evoked increases in redox potential, transforming growth factor- β 1, and fibronectin expression (38). Selective overexpression of MIOX in the absence of hyperglycemia also increases intracellular redox potential, but this effect is attenuated by inhibition of aldehyde reductase, the first enzyme in the GX pathway (38). More recently, pharmacological inhibition of MIOX expression reduced tubular damage and improved kidney function in an animal model of diabetes (40). These findings therefore directly implicate increased MI catabolism in promoting tubular impairment.

As part of our experimental design, we measured MIOX expression in both 4- and 8-wk diabetic kidneys. Interestingly, MIOX expression was decreased in 8-wk diabetic rat kidneys, indicating a temporal relationship with diabetes duration. This decreased expression is consistent with an early study also showing diminished MIOX renal activity in 12-wk diabetic Wistar rats using the same STZ-induced model (37). However, it is in contrast with findings of increased MIOX expression in 5 month diabetic CD1 mice (23) and 12-wk diabetic Sprague-

Dawley rats (21). Both of these studies also utilized STZ induction of diabetes.

It is possible that these differences in MIOX expression are due to species and animal strain differences in genetic background (32). For example, Wistar and Sprague-Dawley rats differ in the rate at which retinopathy develops in STZ-induced diabetes (19). Supporting this possibility in the case of renal susceptibility to diabetes, we reported that 4-wk diabetic male Wistar kidneys subjected to the same STZ-induced experimental regime as described herein displayed significant functional impairment (6). By comparison, symptoms of diabetic nephropathy were notably absent in 4-wk STZ-induced diabetic Sprague-Dawley rats (21).

Collectively, these observations suggest that the Wistar animal strain is comparatively more susceptible to developing nephropathy as a consequence of prolonged hyperglycemia. Changes in MIOX expression are therefore likely disease phase dependent and that ultimately decrease in advanced stages. Interestingly, an acute reduction in MIOX expression has also been reported in acute ischemic renal failure, perhaps indicating the involvement of similar underlying pathophysiological mechanisms (14).

While our findings indicate that MIOX regulates MI renal content, an additional factor within the context of diabetes mellitus is the existence of inosuria. MI is normally filtered from the circulation through the glomeruli and effectively

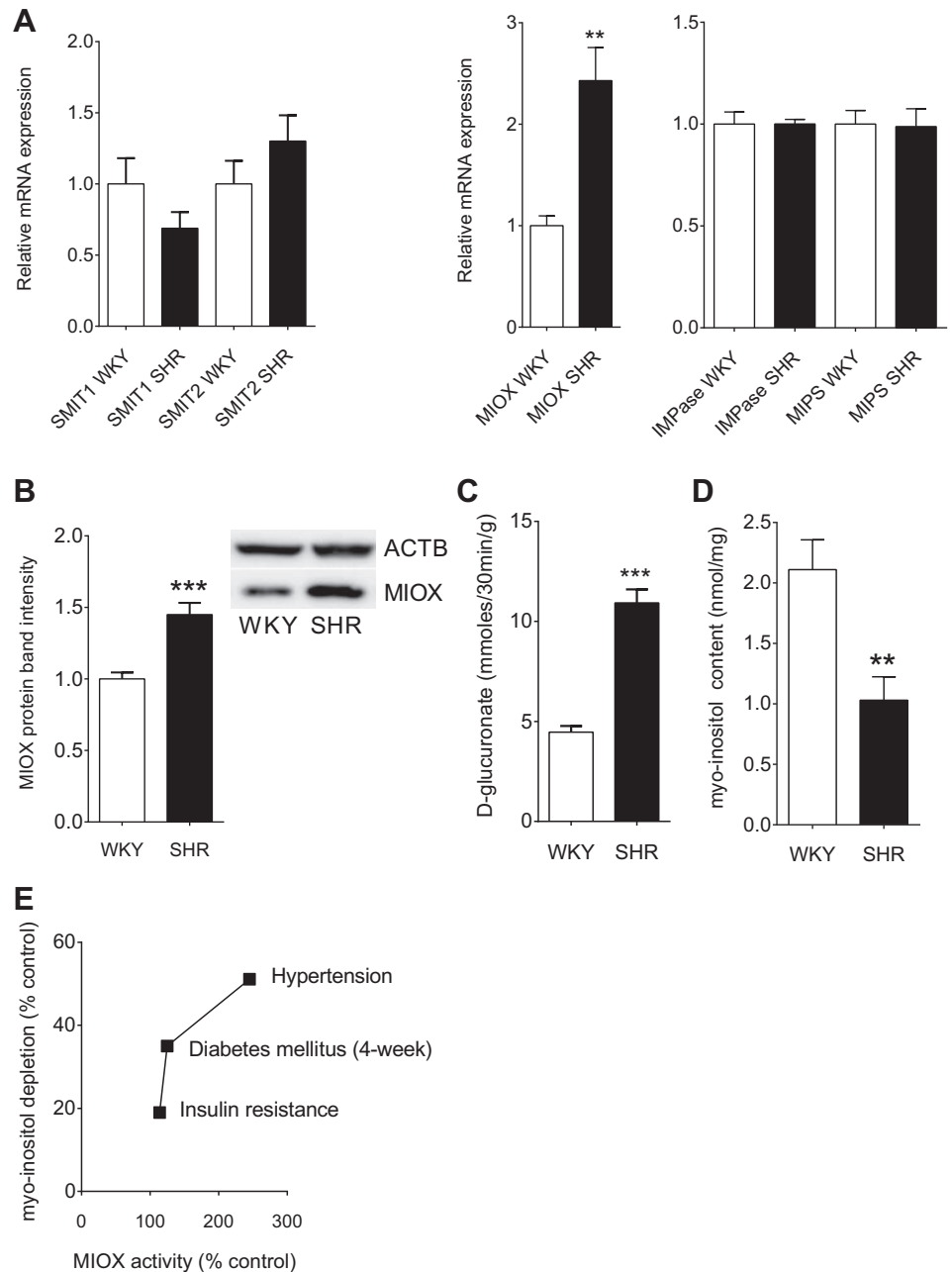


Fig. 6. MI depletion and gene expression profiles in kidneys from spontaneously hypertensive rats (SHR). **A**: mRNA gene expression profiles ($n = 6$ kidneys/group) normalized to the geometric mean of the 4 reference genes, ACTB, HPRT, UBC, and YWHAZ, and expressed as ratios relative to control kidneys. **B**: Western blot analysis of MIOX protein expression. Protein band intensities were normalized to ACTB ($n = 6$ kidneys/group). **C**: MIOX activity measured as mmoles produced over 30 min/g kidney wet wt ($n = 6$ kidneys/group). **D**: intracellular MI concentration normalized to kidney wet wt ($n = 6$ animals/group). **E**: correlation between MI depletion and the fold-percent increase in MIOX activity. Values are means $\pm$ SE. Statistical analysis was performed using Student's unpaired *t*-test. *** $P < 0.001$ vs. Wistar-Kyoto rats (WKY). ** $P < 0.01$ vs. WKY.

reabsorbed predominantly through the proximal tubules. However, in both humans and rodent models of diabetes mellitus there is excessive urinary excretion of MI (18, 25). We were able to replicate inositoria in 4-wk diabetic Wistar kidneys and show it was due to a reduction in MI reabsorption as a consequence of impaired renal function (6). Thus increased MI urinary excretion could also contribute to renal MI intracellular depletion and lower overall body stores. This mechanism could also explain why MI depletion persists in 8-wk diabetic kidneys despite a reduction in MIOX expression.

In summary, renal MI depletion was not limited to diabetes mellitus but was also present in animal models of insulin resistance and hypertension. Our findings indicate that MI depletion could reflect increased catabolism in the proximal tubules through MIOX and the GX pathway, which could

promote oxidative stress. Due to a paucity of research, it remains an open question as to whether MIOX and the GX pathway plays an etiological role in nephropathies associated with human disease. Our findings provide support that further research investigating the role of abnormal MI metabolism in human disease is warranted.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

Author contributions: H.-H.C., A.R.J.P., and K.M.L. provided conception and design of research; H.-H.C., H.-N.C., C.S.W., and S.Y.C. performed experiments; H.-H.C., H.-N.C., and K.M.L. analyzed data; H.-H.C., H.-N.C., C.S.W., and K.M.L. interpreted results of experiments; H.-H.C., H.-N.C., and K.M.L. prepared figures; H.-H.C. and K.M.L. drafted manuscript; H.-H.C., A.R.J.P., and K.M.L. edited and revised manuscript; H.-H.C., A.R.J.P., and K.M.L. approved final version of manuscript.

REFERENCES

- Barski OA, Papusha VZ, Ivanova MM, Rudman DM, Finegold MJ. Developmental expression and function of aldehyde reductase in proximal tubules of the kidney. *Am J Physiol Renal Physiol* 289: F200–F207, 2005.
- Bissonnette P, Coady MJ, Lapointe JY. Expression of the sodium-myo-inositol cotransporter SMIT2 at the apical membrane of Madin-Darby canine kidney cells. *J Physiol* 558: 759–768, 2004.
- Bissonnette P, Lahjouji K, Coady MJ, Lapointe JY. Effects of hyperosmolarity on the Na⁺-myo-inositol cotransporter SMIT2 stably transfected in the Madin-Darby canine kidney cell line. *Am J Physiol Cell Physiol* 295: C791–C799, 2008.
- Bonventre JV. Can we target tubular damage to prevent renal function decline in diabetes? *Semin Nephrol* 32: 452–462, 2012.
- Brown PM, Caradoc-Davies TT, Dickson JM, Cooper GJ, Loomes KM, Baker EN. Purification, crystallization and preliminary crystallographic analysis of mouse myo-inositol oxygenase. *Acta Crystallogr Sect F Struct Biol Cryst Commun* 62: 811–813, 2006.
- Chang HHH, Choong B, Phillips AR, Loomes KM. The diabetic rat kidney mediates inositoluria and selective urinary partitioning of D-chiro-inositol. *Exp Biol Med (Maywood)* 240: 8–14, 2015.
- Chatzigeorgiou A, Halapas A, Kalafatakis K, Kamper E. The use of animal models in the study of diabetes mellitus. *In Vivo* 23: 245–258, 2009.
- Cohen AM, Wald H, Popovtzer M, Rosenmann E. Effect of myo-inositol supplementation on the development of renal pathological changes in the Cohen diabetic (type 2) rat. *Diabetologia* 38: 899–905, 1995.
- Collins A, Foley R, Chavers B, Gilbertson D, Herzog C, Johansen K, Kasiske B, Kutner N, Liu J, St Peter W, Guo H, Gustafson S, Heubner B, Lamb K, Li S, Li S, Peng Y, Qiu Y, Roberts T, Skeans M, Snyder J, Solid C, Thompson B, Wang C, Weinhandl E, Zau D, Arko C, Chen SC, Daniels F, Ebben J, Frazier E, Hanzlik C, Johnson R, Sheets D, Wang X, Forrest B, Constantini E, Everson S, Eggers P, Agodoa L. United States Renal Data System 2011 Annual Data Report: Atlas of Chronic Kidney Disease and End-Stage Renal Disease in the United States. *Am J Kidney Dis* 59: 420, 2012.
- Croze ML, Soulage CO. Potential role and therapeutic interests of myo-inositol in metabolic diseases. *Biochimie* 95: 1811–1827, 2013.
- Gaut JP, Crimmins DL, Ohlendorf MF, Lockwood CM, Griest TA, Brada NA, Hoshi M, Sato B, Hotchkiss RS, Jain S, Ladenson JH. Development of an immunoassay for the kidney-specific protein myo-inositol oxygenase, a potential biomarker of acute kidney injury. *Clin Chem* 60: 747–757, 2014.
- Greene D, Lattimer S, Sima A. Sorbitol, phosphoinositides, and sodium-potassium-ATPase in the pathogenesis of diabetic complications. *N Engl J Med* 316: 599–606, 1987.
- Hankes LV, Politzer WM, Touster O, Anderson L. Myo-inositol catabolism in human pentosurics: the predominant role of the glucuronate-xylulose-pentose phosphate pathway. *Ann NY Acad Sci* 165: 564–576, 1969.
- Hu E, Chen Z, Fredrickson T, Gellai M, Jugus M, Contino L, Spurr N, Sims M, Halsey W, Van Horn S, Mao J, Sathe G, Brooks D. Identification of a novel kidney-specific gene downregulated in acute ischemic renal failure. *Am J Physiol Renal Physiol* 279: F426–F439, 2000.
- Hultstrom M. Development of structural kidney damage in spontaneously hypertensive rats. *J Hypertens* 30: 1087–1091, 2012.
- Jha V, Garcia-Garcia G, Iseki K, Li Z, Naicker S, Plattner B, Saran R, Wang A, Yang CW. Chronic kidney disease: global dimension and perspectives. *Lancet* 382: 260–272, 2013.
- Karlen Y, McNair A, Perseguers S, Mazza C, Mermod N. Statistical significance of quantitative PCR. *BMC Bioinformatics* 8: 131, 2007.
- Kawa JM, Przybylski R, Taylor CG. Urinary chiro-inositol and myo-inositol excretion is elevated in the diabetic db/db mouse and streptozotocin diabetic rat. *Exp Biol Med (Maywood)* 228: 907–914, 2003.
- Kern TS, Miller CM, Tang J, Du Y, Ball SL, Berti-Matera L. Comparison of three strains of diabetic rats with respect to the rate at which retinopathy and tactile allodynia develop. *Mol Vision* 16: 1629–1639, 2010.
- Lahjouji K, Aouameur R, Bissonnette P, Coady MJ, Bichet DG, Lapointe JY. Expression and functionality of the Na⁺/myo-inositol cotransporter SMIT2 in rabbit kidney. *Biochim Biophys Acta* 1768: 1154–1159, 2007.
- Lu Y, Liu C, Miao X, Xu K, Wu X, Liu C. Increased expression of myo-inositol oxygenase is involved in the tubulointerstitial injury of diabetic nephropathy. *Exp Clin Endocrinol Diabetes* 117: 257–265, 2009.
- Mennuni S, Rubattu S, Pierelli G, Tocci G, Fofi C, Volpe M. Hypertension and kidneys: unraveling complex molecular mechanisms underlying hypertensive renal damage. *J Hum Hypertens* 28: 74–79, 2014.
- Nayak B, Kondeti V, Xie P, Lin S, Viswakarma N, Raparia K, Kanwar Y. Transcriptional and post-translational modulation of myo-inositol oxygenase by high glucose and related pathobiological stresses. *J Biol Chem* 286: 27594–28205, 2011.
- Nayak B, Xie P, Akagi S, Yang Q, Sun L, Wada J, Thakur A, Danesh FR, Chugh SS, Kanwar YS. Modulation of renal-specific oxidoreductase/myo-inositol oxygenase by high-glucose ambience. *Proc Natl Acad Sci USA* 102: 17952–17957, 2005.
- Ostlund RE Jr, McGill JB, Herskowitz I, Kipnis DM, Santiago JV, Sherman WR. D-chiro-inositol metabolism in diabetes mellitus. *Proc Natl Acad Sci USA* 90: 9988–9992, 1993.
- Parthasarathy LK, Seelan RS, Tobias C, Casanova MF, Parthasarathy RN. Mammalian inositol 3-phosphate synthase: its role in the biosynthesis of brain inositol and its clinical use as a psychoactive agent. *Subcell Biochem* 39: 293–314, 2006.
- Prabhu KS, Arner RJ, Vunta H, Reddy CC. Up-regulation of human myo-inositol oxygenase by hyperosmotic stress in renal proximal tubular epithelial cells. *J Biol Chem* 280: 19895–19901, 2005.
- Prasad GV. Metabolic syndrome and chronic kidney disease: current status and future directions. *World J Nephrol* 3: 210–219, 2014.
- Raccach D, Coste T, Cameron N, Dufayet D, Vague P, Hohman T. Effect of the aldose reductase inhibitor tolrestat on nerve conduction velocity, Na/K ATPase activity, and polyols in red blood cells, sciatic nerve, kidney cortex, and kidney medulla of diabetic rats. *J Diabetes Complications* 12: 154–162, 1998.
- Reddy CC, Swan JS, Hamilton GA. myo-Inositol oxygenase from hog kidney. I. Purification and characterization of the oxygenase and of an enzyme complex containing the oxygenase and D-glucuronate reductase. *J Biol Chem* 256: 8510–8518, 1981.
- Small DM, Coombes JS, Bennett N, Johnson DW, Gobe GC. Oxidative stress, anti-oxidant therapies and chronic kidney disease. *Nephrology (Carlton)* 17: 311–321, 2012.
- Soler MJ, Riera M, Battle D. New experimental models of diabetic nephropathy in mice models of type 2 diabetes: efforts to replicate human nephropathy. *Exp Diabetes Res* 2012: 616313, 2012.
- Tesch GH, Allen TJ. Rodent models of streptozotocin-induced diabetic nephropathy. *Nephrology* 12: 261–266, 2007.
- Touster O. Essential pentosuria and the glucuronate-xylulose pathway. *Fed Proc* 19: 977–983, 1960.
- Vallon V, Thomson S. Renal function in diabetic disease models: the tubular system in the pathophysiology of the diabetic kidney. *Annu Rev Physiol* 74: 351–375, 2012.
- Walker CS, Li X, Whiting L, Glyn-Jones S, Zhang S, Hickey AJ, Sewell MA, Ruggiero K, Phillips AR, Kraegen EW, Hay DL, Cooper GJ, Loomes KM. Mice lacking the neuropeptide alpha-calcitonin gene-related peptide are protected against diet-induced obesity. *Endocrinology* 151: 4257–4269, 2010.
- Whiting PH, Palmano KP, Hawthorne JN. Enzymes of myo-inositol and inositol lipid metabolism in rats with streptozotocin-induced diabetes. *Biochem J* 179: 549–553, 1979.
- Xie P, Sun L, Oates P, Srivastava S, Kanwar Y. Pathobiology of renal-specific oxidoreductase/myo-inositol oxygenase in diabetic nephropathy: its implications in tubulointerstitial fibrosis. *Am J Physiol Renal Physiol* 298: F1396–F1404, 2010.
- Yang Q, Dixit B, Wada J, Tian Y, Wallner EI, Srivastava SK, Kanwar YS. Identification of a renal-specific oxido-reductase in newborn diabetic mice. *Proc Natl Acad Sci USA* 97: 9896–9901, 2000.
- Zhan M, Usman IM, Sun L, Kanwar YS. Disruption of renal tubular mitochondrial quality control by myo-inositol oxygenase in diabetic kidney disease. *J Am Soc Nephrol* 26: 1304–1321, 2015.